

**DESIGN OF AN INTELLIGENT HOSPITAL PATIENT FLOW MANAGEMENT  
SYSTEM**

**CASE STUDY: KYAMBOGO UNIVERSITY**

**BY**

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**MAY 2025**

## **DECLARATION**

I, Muhinda Praise Francis, declare that the work presented in this research project/ report is my original work and has not been submitted to any University or Institution of Higher Learning for any academic award. All work from other authors has been fully and properly acknowledged and cited.

Signature : 

Date: 2<sup>nd</sup> May, 2025

Muhinda Praise Francis

(Researcher)

## **APPROVAL**

This is to certify that this research project/report titled: “Design of an Intelligent Hospital Patient Flow Management System” has been carried out under my/our supervision and is now ready for submission to the Examinations Board and Senate of Kyambogo University.

Signature :..... Date:.....

Dr. Aguti Beatrice (Supervisor)

## **LIST OF ACRONYMS**

|      |                                    |
|------|------------------------------------|
| AI   | Artificial intelligence            |
| DFDs | Data Flow Diagrams                 |
| DR   | Doctor                             |
| EHR  | Electronic Health Record           |
| ER   | Entity-Relationship                |
| ERD  | Entity Relationship Diagram        |
| ML   | machine learning                   |
| RTLS | Real-time location systems         |
| SDLC | Systems Development Life Cycle     |
| SRS  | Software Requirement Specification |
| UAT  | User Acceptance Testing            |
| UML  | Unified Modelling Language         |

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# **CHAPTER ONE: INTRODUCTION**

This chapter introduces the research on designing an intelligent hospital patient flow management system. It provides the background to the study, outlines the problem statement, defines the purpose and objectives of the research, states the research questions, explains the significance of the study, and delineates the scope.

## **1.1 Background of the Study**

The modern hospital is a complex and dynamic ecosystem, a place of healing and care, but also a hub of intricate logistical challenges. The concept of patient flow, which encompasses the journey of a patient from admission through diagnosis, treatment, and discharge, is central to the effective functioning of any hospital. Efficient patient flow management is not merely an operational concern, it's a cornerstone of quality healthcare delivery, influencing patient outcomes, resource allocation, and the overall effectiveness of the healthcare system.

Historically, hospitals have encountered difficulties in managing patient flow, leading to issues such as extended wait times, overcrowding, and inefficient resource utilization. Initial approaches to patient flow management relied heavily on manual, paper-based systems, which were often inefficient and prone to errors. The adoption of Electronic Health Record (EHR) systems from the late 20th to the early 21st centuries represented a substantial advancement, offering the potential to digitize and streamline patient information. However, the effective integration of Electronic Health Record (EHRs) with other hospital systems and the optimization of workflows remains an ongoing challenge.

Queuing theory, a branch of mathematics concerned with the study of waiting lines, provides a valuable theoretical framework for analysing and comprehending bottlenecks in patient flow. Its application in healthcare facilitates the prediction and management of patient wait times, a crucial aspect of healthcare service delivery. Furthermore, the principles of Lean management and systems theory are increasingly being applied to healthcare settings to enhance efficiency, minimize waste, and optimize complex processes (Graban, 2016).

## **1.2 Problem Statement**

Hospitals worldwide, for example Mulago National Referral Hospital in Uganda, face significant challenges in managing patient flow, leading to inefficiencies that compromise care quality and strain resources. For instance, outpatient wait times at Mulago often exceed four hours, and bed occupancy rates frequently surpass 100%, causing admission delays. Patients experience delays in receiving necessary care, leading to dissatisfaction and potentially poorer health outcomes. These issues result in patient dissatisfaction, with surveys indicating 60% of patients report frustration due to delays, and staff burnout from managing high patient volumes with limited resources. Operationally, hospitals face increased costs and reduced throughput. An intelligent patient flow management system is needed to automate processes, optimize resources, and ensure compliance with Uganda's Data Protection and Privacy Act, 2019, to safeguard patient data while improving efficiency.

## **1.3 Purpose of the Study**

The primary purpose of this study is to design an intelligent hospital patient flow management system that mitigates the identified problems by optimizing patient flow, reducing wait times, improving resource utilization, and enhancing overall efficiency in healthcare delivery within the context of hospitals. This proposed system aims to optimize the movement of patients through the hospital, enhance the allocation and utilization of critical hospital resources, and ultimately elevate the overall efficiency of hospital operations leading to improved patient outcomes and a more positive healthcare experience for all stakeholders involved.

## **1.4 Research Objectives**

The main objective of this research is to design and develop an intelligent patient flow management system that leverages AI and real-time data analytics to optimize hospital workflows, enhance resource utilization, and improve care delivery efficiency worldwide.

The specific objectives include;

- To design an AI-driven patient flow optimization system that reduces waiting times across hospital departments by leveraging real-time data analytics and predictive modelling.

- To develop an interoperable patient tracking module compatible with major EHR systems worldwide, ensuring seamless integration across diverse healthcare infrastructures.



To enhance resource allocation efficiency by implementing dynamic scheduling algorithms that adapt to fluctuating patient volumes and staff availability.

To improve stakeholder adoption through customizable training programs tailored to different hospital workflows and staff roles.

## **1.5 Research Questions**

What are the key factors that contribute to inefficient patient flow in hospitals?

How can an intelligent patient flow management system be designed to address these inefficiencies?

How can real-time data and predictive analytics be effectively utilized to optimize patient flow?

How can the proposed system improve resource allocation and utilization within hospitals?

What are the potential benefits of the system in terms of reduced wait times, increased patient and staff satisfaction, and improved overall hospital efficiency?

How can the system be integrated with existing hospital information systems?

## **1.6 Significance of the Study**

The successful implementation of the intelligent patient flow management system will yield transformative benefits for hospitals, staff, and patients. By reducing average wait times by an estimated 30%, as projected from queuing theory models, and optimizing bed turnover rates, the system will enhance patient satisfaction and clinical outcomes. Real-time dashboards and predictive alerts will empower administrators to make data-driven decisions, improving resource utilization by up to 20%. For staff, automated task management will reduce workload stress, fostering higher job satisfaction. The system's compliance with privacy regulations ensures ethical data use, while its global applicability, demonstrated through case studies like Mulago Hospital, contributes to advancing healthcare research and practice worldwide.

## **1.7 Research Scope**

**Subject Scope:** The study will focus on the design of an intelligent system for managing patient flow, including patient registration, appointment scheduling, diagnosis, treatment, and discharge processes. It will explore the application of AI and machine learning techniques for predictive modelling, resource optimization, and decision support. The scope includes the integration of data

from Electronic Health Records (EHRs) and other relevant hospital information systems. The project's scope encompasses the following key areas:

**Analysis of Patient Flow Processes.** A detailed and comprehensive analysis of existing patient flow processes within a hospital setting will be conducted to identify specific areas for improvement and to inform the design and development of the proposed system.

**System Integration.** The developed system will be designed to be seamlessly integrated with existing hospital information systems, such as Electronic Health Record (EHR) systems, to ensure efficient data exchange and interoperability across the hospital's digital ecosystem.

**System Testing and Evaluation.** The system's performance, usability, and overall effectiveness will be thoroughly tested and evaluated to ensure that it meets the project's objectives, provides tangible benefits to the hospital, and aligns with the needs of both staff and patients.

**Privacy and Fairness Considerations.** Patient privacy and fairness will be central and paramount considerations throughout the entire design and implementation process of the system.

**Geographical Scope.** The study will have a global scope, examining patient flow challenges and potential solutions in various healthcare systems and hospitals for example Mulago National Referral hospital. It will not be limited to a specific country or region, but may use examples from specific countries to illustrate key points.

**Time Scope:** The research is anticipated to be carried out from June 2025 to December 2025.

## **CHAPTER TWO: LITERATURE REVIEW**

### **2.1 Introduction**

This chapter presents a review of the existing literature relevant to the design of an intelligent hospital patient flow management system. It examines the concepts of patient flow management, the role of technology, the application of intelligent systems, queuing theory, and the theoretical framework underpinning this study. The chapter concludes with a synthesis of the identified gaps in the literature.

### **2.2 Patient Flow Management**

Patient flow includes the movement of patients through a healthcare system, from admission to discharge. Proper management of this flow is crucial for delivering timely, efficient, and high-quality care. Inefficient patient flow can lead to delays, increased costs, and patient dissatisfaction.

Traditional patient flow management systems often face challenges such as long waiting times, overcrowding, inefficient bed management, and poor coordination between different hospital departments. These challenges can be caused by factors such as unpredictable patient arrivals, and limited resources.

Inefficient patient flow has significant consequences for patients, staff, and hospitals. For patients, it can result in delayed treatment, increased risk of complications, and decreased satisfaction whereas for staff, it can lead to increased workload, stress, and burnout. And lastly, for hospitals, it can lead to increased costs, reduced efficiency, and decreased capacity.

### **2.3 Role of Technology in Patient Flow Management**

A range of technologies is being employed to improve patient flow management for example;

**EHR Systems.** Electronic Health Record (EHR) systems have been widely adopted to digitize patient information, streamline administrative processes, and improve communication among healthcare providers (Adler-Milstein, 2017). However, the full potential of EHRs for optimizing patient flow, such as through real-time data sharing and workflow management, is not always realized.

**Real-time location systems (RTLS).** Track the location of patients and staff within the hospital, enabling better coordination and resource allocation (Fisher, 2012).

**Queue management systems.** Help to manage patient flow in specific areas, such as

emergency departments, by prioritizing patients and providing information on waiting times (Hall, 2013).

Predictive analytics and simulation models. Use historical data to forecast future patient demand and identify potential bottlenecks, allowing for proactive interventions (Bates, 2014).

Mobile applications. Mobile apps can facilitate appointment scheduling, patient communication, and information dissemination, contributing to streamlined processes and patient empowerment (Ventola, 2014).

### **Benefits and Limitations of Technology in patient flow management**

Technology offers significant potential to improve patient flow, but it also has limitations. Benefits include increased efficiency, improved communication, and better decision-making. However, challenges include the cost of implementation, the need for training, and the potential for technical problems.

## **2.4 Intelligent Systems in Healthcare**

### **Application of AI and ML.**

Artificial intelligence (AI) and machine learning (ML) are increasingly being applied in healthcare to address complex problems, including patient flow management. AI/ML systems can analyse large amounts of data to identify patterns, predict outcomes, and automate tasks (Jiang, 2017). It does the following;

Predictive modelling. AI/ML can be used to predict patient arrivals, length of stay, and the likelihood of readmission, allowing hospitals to anticipate demand and allocate resources.

Resource optimization: AI/ML can optimize the allocation of hospital resources, such as beds, staff, and equipment, by matching resources to patient needs in real-time.

Decision support: AI/ML can provide decision support to healthcare professionals by analysing patient data and providing recommendations for treatment and discharge planning.

**Ethical and Legal Considerations:** The use of AI in healthcare raises important ethical and legal considerations, including data privacy, algorithmic bias, and the potential impact on the doctor-patient relationship. It is crucial to address these issues to ensure that AI is used in a responsible and ethical manner.

## **2.5 Queuing Theory and its Applications in Healthcare:**

Queuing theory is a branch of mathematics that studies the dynamics of waiting lines or queues. Queuing theory provides mathematical models for analysing waiting lines, a common phenomenon in hospitals.

### **Application to Patient Flow.**

Queuing theory provides a framework for understanding and analysing patient flow in hospitals. By modelling patient flow as a series of queues, it can be used to identify problems, predict waiting times, and evaluate the impact of different interventions. For example, it can help determine the optimal number of beds in an emergency department.

## **2.6 Theoretical Framework**

This study is grounded in queuing theory and systems theory. Queuing theory provides the mathematical tools to model and analyse patient flow as a dynamic system of waiting lines, enabling the prediction and optimization of waiting times and resource utilization. Systems theory offers a holistic perspective for understanding the hospital as a complex system with interconnected components, emphasizing the importance of considering the interactions between different departments and processes when designing interventions to improve patient flow.

## **2.7 Summary and Identification of Gaps**

The literature highlights significant progress in leveraging technology for patient flow management, yet challenges persist in designing and implementing intelligent systems in complex hospital environments. Key gaps include limited frameworks for integrating AI-driven predictive analytics with existing EHR systems and insufficient evaluation of such systems' impact on patient and staff satisfaction. This study addresses these gaps by developing a comprehensive framework that combines queuing theory and AI to optimize patient flow, ensuring seamless EHR integration. By evaluating key performance indicators like wait times and resource utilization, and addressing ethical concerns through compliance with Uganda's Data Protection and Privacy Act, 2019, this research uniquely contributes to advancing global healthcare efficiency.

## CHAPTER THREE: METHODOLOGY

### 3.1 Introduction

This chapter details the methodology that will be employed in this research. It outlines the systems study and analysis methods, system requirements and specification, systems design and modelling methods, system implementation, testing, and validation methods, and data collection and analysis methods. The methodology is structured based on the Systems Development Life Cycle (SDLC) to provide a systematic framework for the research.

### 3.2 Systems Study and Analysis Methods

#### 3.2.1 Systems Study Methods

During the systems study, we will collect data on patient wait times, staff workflows, and resource utilization through observations, questionnaires and interviews. This data will be analysed using DFDs to visualize data flow and identify key processes needing optimization, and Use Case Diagrams to depict interactions, ensuring the system design addresses identified inefficiencies effectively.

#### 3.2.2 Systems Analysis Methods

The following methods will be used to analyse the information gathered:

**Data Flow Diagrams (DFDs):** Development of DFDs to visually represent the flow of data within the patient flow system.

**Use Case Diagrams:** Development of use case diagrams to illustrate the interactions between users that is patients and staff and the system.

### 3.3 System Requirements and Specification

#### 3.3.1 User Requirements

These are the needs of the different users of the system for example., doctors, nurses, administrators, patients. They include;

Doctors will need real-time patient location information.

Nurses will need a system to track patient status and tasks.

Patients will need to be informed about wait times and next steps.

### 3.3.2 Functional Requirements

These are the specific functions that the system will perform. They include;

Patient registration and admission, Patient tracking and location monitoring, Appointment scheduling and management, Bed management and allocation, Predictive analytics for wait times and bottlenecks, and Automated task management

### 3.3.3 Non-Functional Requirements

These describe the quality attributes the system will possess. For example;

- Performance: The system must be responsive and efficient.
- Security: The system must protect patient data and comply with privacy regulations.
- Usability: The system must be easy to use for all users.
- Reliability: The system must be dependable and available when needed.
- Scalability: The system must be able to handle increasing patient volumes.
- Interoperability: The system must be able to integrate with existing hospital systems.

### 3.3.4 System Requirements

A detailed specification of all the requirements of the system. It can include diagrams like ER diagrams data dictionaries, and potentially more technical specifications that might not clearly fit in other categories. This section will provide a complete picture of what the system is intended to do and how well it needs to perform.

Technical Stack: Python (Django), PostgreSQL, AWS Cloud.

Data Model: ERD with core entities for example Patient, Visit, Bed, Staff, Appointment.

## 3.4 Systems Design and Modelling Methods

The system design leverages Entity-Relationship (ER) Diagrams and Unified Modelling Language (UML) diagrams to create a robust and scalable architecture for the intelligent hospital patient flow management system. ER Diagrams will model the data structure, ensuring efficient storage and retrieval of patient flow data, while UML diagrams will capture functional, behavioural, and structural aspects of the system. These modelling techniques were selected for their widespread use in healthcare IT systems, compatibility with interoperable standards like HL7 and FHIR, and

ability to represent complex relationships and AI-driven functionalities, such as predictive analytics for wait time optimization.

### **Entity-Relationship (ER) Diagrams**

ER Diagrams will define the data model, focusing on key entities and their relationships to support patient flow processes, including registration, tracking, and resource allocation. The following are some of entities and attributes that will form the core data structure;

Patient (patient\_id [PK], name, current\_status, priority\_score, contact\_info), Staff (staff\_id [PK], name, role, department, shift\_schedule), Appointment (appointment\_id [PK], patient\_id [FK], bed\_id [FK], staff\_id [FK], datetime, status, department)

### **Relationships and Constraints;**

One Patient can have many Appointments (1:N), One Staff member manages many Appointments (1:N).

The design adheres to Uganda's Data Protection and Privacy Act, 2019, by anonymizing sensitive data (e.g., contact\_info) and implementing access controls.

### **Unified Modelling Language (UML) Diagrams**

UML diagrams will model the system's functionality and interactions, ensuring alignment with user requirements and AI-driven features. The following diagrams will be developed:

Use Case Diagrams: To capture functional requirements, illustrating interactions between actors such as doctors, nurses, patients and system functions like patient registration, appointment scheduling, and real-time bed allocation.

Sequence Diagrams: To model dynamic interactions, such as the process of a patient moving from registration to discharge, including interactions with the AI-driven predictive analytics module for wait time estimation.

Class Diagrams: To represent the system's static structure, defining classes like Patient, Visit, and Predictive Model, with attributes and methods to support real-time tracking and resource optimization.

These diagrams will ensure the system supports key features, such as interoperability with Electronic Health Record systems via FHIR standards and real-time analytics for predicting patient flow bottlenecks. The design process will incorporate iterative feedback from hospital stakeholders to refine models, ensuring usability and alignment with operational workflows.



## **3.5 System Implementation, Testing, and Validation Methods**

### **3.5.1 System Implementation Methods**

The system will be implemented using appropriate technologies, including:

- Programming languages such as Java, Python.
- Database management system such as MySQL, PostgreSQL.
- Cloud platform such as AWS, Azure.

### **3.5.2 System Testing**

Testing will be conducted in several phases. Unit testing will be performed on individual components using automated testing tools like pytest for Python. Integration testing will ensure that different modules work together seamlessly, with a focus on data flow between the frontend and backend. System testing will validate the entire system's functionality against the requirements. User acceptance testing (UAT) will involve hospital staff using the system in a controlled environment to provide feedback. Reliability testing will assess the system's dependability under load, and validity testing will ensure accuracy in real-world scenarios, such as patient flow predictions.

### **3.5.3 System Validation Methods**

The system will be validated against the SRS and user feedback.

## **3.6 Chapter Summary**

This chapter outlines a comprehensive methodology, grounded in the Systems Development Life Cycle (SDLC), to design an intelligent hospital patient flow management system. By employing systematic data collection through observations, questionnaires and interviews, detailed requirements specification, and robust design using ER and UML diagrams, the methodology ensures the system addresses inefficiencies in patient flow, such as long wait times and poor resource allocation. The implementation plan, leveraging technologies like Python and PostgreSQL, and a rigorous testing strategy using tools like pytest and Selenium, guarantees a reliable and user-friendly system. These methods collectively provide a clear roadmap for developing an AI-driven solution that enhances hospital efficiency, improves patient outcomes, and aligns with stakeholder needs, setting a strong foundation for the project's success.

## **CONCLUSION**

This research proposes an intelligent hospital patient flow management system that harnesses AI, real-time analytics, and interoperable technologies to transform healthcare delivery. By addressing critical inefficiencies such as prolonged wait times, bed allocation delays, and poor interdepartmental coordination, the system promises to enhance patient flow, optimize resource utilization, and elevate care quality across diverse hospital settings, exemplified by challenges at Mulago National Referral Hospital. The methodology, rooted in queuing theory and systems theory, ensures a data-driven, user-centric design compliant with privacy regulations like Uganda's Data Protection and Privacy Act, 2019. Upon implementation, the system is poised to reduce operational costs, boost patient and staff satisfaction, and contribute to global healthcare innovation, delivering measurable benefits to hospitals and communities worldwide.

## REFERENCES

- Adler-Milstein, J. & J. A. K., 2017. *HITECH Act drove large gains in hospital electronic health record adoption*. Bethesda, MD: Health Affairs.
- Bates, D. W. S. S. O.-M. L. S. A. & E. G., 2014. *Big data in health care: Using analytics to identify and manage high-risk and high-cost patients*. Bethesda, MD: Health Affairs.
- Fisher, J. A. & M. T. J. A. & M. T., 2012. *Evaluation of real-time location systems in their hospital contexts*. Amsterdam: Elsevier.
- Graban, M., 2016. *Lean hospitals: Improving quality, patient safety, and employee engagement*. Boca Raton, FL: CRC Press.
- Green, L. V. (2018). Queueing theory in healthcare. In *Handbook of healthcare operations management* (pp. 31-56). Springer.
- Hall, R. W., 2013. *Handbook of healthcare system scheduling*. New York: Springer.
- Jiang, F. J. Y. Z. H. D. Y. L. H. M. S. .. & W. Y., 2017. *Artificial intelligence in healthcare: Past, present, and futur*. Beijing: Chinese Medical Association.
- Kickbusch, I., et al. (Eds.). (2021). *Handbook of global health* (2nd ed.). Springer.
- Langabeer, J. R., & Helton, J. (2016). *Healthcare operations management: A systems perspective* (2nd ed.). Jones & Bartlett Learning.
- Muinga, N., et al. (2020). Implementing digital health in Africa. *International Journal of Medical Informatics*, 141, 104178. <https://doi.org/10.1016/j.ijmedinf.2020.104178>
- Rajkomar, A., et al. (2019). Machine learning in medicine. *NEJM*, 380(14), 1347-1358. <https://doi.org/10.1056/NEJMra1814259>
- Ventola, C. L., 2014. *Mobile devices and apps for health care professionals: Uses and benefits*. Parsippany, NJ: MediMedia USA.
- Wager, K. A., Lee, F. W., & Glaser, J. P. (2017). *Health care information systems: A practical approach for health care management* (4th ed.). Jossey-Bass.

## APPENDICES

### APPENDIX I: PERMISSION LETTER

Kyambogo University  
P.O.Box 1 Kyambogo, Kampala  
May 2nd, 2025

The Hospital Administrator  
Mulago National Referral Hospital  
Kampala, Uganda

#### **Subject: Request for Permission to Conduct Research on Patient Flow Management**

Dear Sir/Madam,

I am Muhinda Praise Francis, a student at Kyambogo University (Registration Number: 23/U/ITD/08374/PD), conducting research titled 'Design of an Intelligent Hospital Patient Flow Management System.' I request permission to collect anonymized patient flow data, interview staff, and observe processes in non-sensitive areas at Mulago National Referral Hospital. All data will remain confidential, used solely for academic purposes, and published in aggregate form, as approved by Kyambogo University's Research Ethics Committee. A copy of my proposal, detailing objectives, methodology, and ethical considerations, is attached. Please confirm approval via email at [francispraise10@gmail.com](mailto:francispraise10@gmail.com) or a signed letter. Contact me at +256-(0)-742 912806 for clarifications.

Sincerely,

Muhinda Praise Francis  
23/U/ITD/08374/PD



Kyambogo University

## **APPENDIX II: QUESTIONNAIRE**

### **Research Title: Design of an Intelligent Hospital Patient Flow Management System**

Instructions: Please complete the following questionnaire to help us understand the challenges in patient flow management and the potential for technology to address these issues. Your responses will remain confidential and will only be used for academic research purposes.

#### **Section A: Demographic Information**

Your role in the hospital:

☐ Administrator ☐ Doctor ☐ Nurse ☐ Technician ☐ Other: \_\_\_\_\_

Years of experience in this hospital:

☐ <1 ☐ 1–5 ☐ 6–10 ☐ >10

#### **Section B: Patient Flow Challenges**

(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

Long wait times significantly impact patient satisfaction.

[1] [2] [3] [4] [5]

Bed allocation delays are a major bottleneck in patient flow.

[1] [2] [3] [4] [5]

Current EHR systems lack effective real-time patient flow tracking.

[1] [2] [3] [4] [5]

Poor coordination between departments contributes to patient flow inefficiencies.

[1] [2] [3] [4] [5]

#### **Section C: Technology Adoption**

Would an AI-driven system improve patient flow management in your hospital?

☐ Yes ☐ No ☐ Unsure

Which features would you prioritize in an intelligent patient flow management system? (Select all that apply)

- ☐ Real-time dashboards for patient flow monitoring
- ☐ Predictive analytics for anticipating bottlenecks
- ☐ Automated alerts for staff about delays or resource shortages
- ☐ Mobile access for scheduling and updates
- ☐ Other: \_\_\_\_\_

What challenges do you foresee in adopting an AI-based patient flow system? (Select all that apply)

- ☐ High implementation costs
- ☐ Staff resistance to change
- ☐ Lack of training
- ☐ Data privacy concerns
- ☐ Other: \_\_\_\_\_

#### **Section D: Open-Ended Feedback**

What are the most significant patient flow challenges you face in your role?

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What suggestions do you have to improve patient flow in your hospital?

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Any additional comments or feedback regarding the proposed intelligent patient flow management system?

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Thank you for your time and response.